PG No. 32 O_{h} $m\bar{3}m$ [cubic] (internal axial dipole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_{0}^{(1,1)}[g](A_{1u})$

** symmetry

1

** expression

$$\frac{\sqrt{3}G_x x}{3} + \frac{\sqrt{3}G_y y}{3} + \frac{\sqrt{3}G_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_{1,1}^{(1,-1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(1,-1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{1,3}^{(1,-1)}[g](T_{1g})$

** symmetry

x

y

z

** expression

G_x

G_y

G_z

$\vec{\mathbb{G}}_{1,1}^{(1,1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(1,1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{1,3}^{(1,1)}[g](T_{1g})$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{10}G_x\left(2x^2 - y^2 - z^2\right)}{10} + \frac{3\sqrt{10}G_y xy}{10} + \frac{3\sqrt{10}G_z xz}{10}$$

$$\frac{3\sqrt{10}G_x xy}{10} - \frac{\sqrt{10}G_y\left(x^2 - 2y^2 + z^2\right)}{10} + \frac{3\sqrt{10}G_z yz}{10}$$

$$\frac{3\sqrt{10}G_x xz}{10} + \frac{3\sqrt{10}G_y yz}{10} - \frac{\sqrt{10}G_z\left(x^2 + y^2 - 2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,0)}[g](T_{1u})$, $\vec{\mathbb{Q}}_{1,2}^{(1,0)}[g](T_{1u})$, $\vec{\mathbb{Q}}_{1,3}^{(1,0)}[g](T_{1u})$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{2}G_y z}{2} - \frac{\sqrt{2}G_z y}{2}$$

$$-\frac{\sqrt{2}G_x z}{2} + \frac{\sqrt{2}G_z x}{2}$$

$$\frac{\sqrt{2}G_x y}{2} - \frac{\sqrt{2}G_y x}{2}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_{2,1}^{(1,0)}[g](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(1,0)}[g](E_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{6}G_x yz}{2} - \frac{\sqrt{6}G_y xz}{2}$$

$$\frac{\sqrt{2}G_x yz}{2} + \frac{\sqrt{2}G_y xz}{2} - \sqrt{2}G_z xy$$

$\vec{\mathbb{Q}}^{(1,0)}_{2,1}[g](T_{2g})$, $\vec{\mathbb{Q}}^{(1,0)}_{2,2}[g](T_{2g})$, $\vec{\mathbb{Q}}^{(1,0)}_{2,3}[g](T_{2g})$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_x\,(y-z)\,(y+z)}{2} - \frac{\sqrt{2}G_y xy}{2} + \frac{\sqrt{2}G_z xz}{2}$$

$$\frac{\sqrt{2}G_x xy}{2} - \frac{\sqrt{2}G_y\,(x-z)\,(x+z)}{2} - \frac{\sqrt{2}G_z yz}{2}$$

$$-\frac{\sqrt{2}G_x xz}{2} + \frac{\sqrt{2}G_y yz}{2} + \frac{\sqrt{2}G_z\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{G}}^{(1,-1)}_{2,1}[g](E_u)$, $\vec{\mathbb{G}}^{(1,-1)}_{2,2}[g](E_u)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{6}G_x x}{6} - \frac{\sqrt{6}G_y y}{6} + \frac{\sqrt{6}G_z z}{3}$$

$$\frac{\sqrt{2}G_x x}{2} - \frac{\sqrt{2}G_y y}{2}$$

$\vec{\mathbb{G}}^{(1,1)}_{2,1}[g](E_u)$, $\vec{\mathbb{G}}^{(1,1)}_{2,2}[g](E_u)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{21}G_x x\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}G_y y\left(x^2+y^2-4z^2\right)}{14} - \frac{\sqrt{21}G_z z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$\frac{\sqrt{7}G_x x\left(3x^2-7y^2-2z^2\right)}{14} + \frac{\sqrt{7}G_y y\left(7x^2-3y^2+2z^2\right)}{14} + \frac{5\sqrt{7}G_z z\,(x-y)\,(x+y)}{14}$$

$\vec{\mathbb{G}}^{(1,-1)}_{2,1}[g](T_{2u})$, $\vec{\mathbb{G}}^{(1,-1)}_{2,2}[g](T_{2u})$, $\vec{\mathbb{G}}^{(1,-1)}_{2,3}[g](T_{2u})$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_y z}{2} + \frac{\sqrt{2}G_z y}{2}$$

$$\frac{\sqrt{2}G_x z}{2} + \frac{\sqrt{2}G_z x}{2}$$

$$\frac{\sqrt{2}G_x y}{2} + \frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,1)}[g](T_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(1,1)}[g](T_{2u})$, $\vec{\mathbb{G}}_{2,3}^{(1,1)}[g](T_{2u})$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{7}G_x xyz}{7} - \frac{\sqrt{7}G_y z\left(x^2 - 4y^2 + z^2\right)}{7} - \frac{\sqrt{7}G_z y\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$\frac{\sqrt{7}G_x z\left(4x^2 - y^2 - z^2\right)}{7} + \frac{5\sqrt{7}G_y xyz}{7} - \frac{\sqrt{7}G_z x\left(x^2 + y^2 - 4z^2\right)}{7}$$

$$\frac{\sqrt{7}G_x y\left(4x^2 - y^2 - z^2\right)}{7} - \frac{\sqrt{7}G_y x\left(x^2 - 4y^2 + z^2\right)}{7} + \frac{5\sqrt{7}G_z xyz}{7}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_{3}^{(1,-1)}[g](A_{2g})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_x yz + G_y xz + G_z xy$$

$\vec{\mathbb{G}}_{3}^{(1,1)}[g](A_{2g})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}G_x yz\left(6x^2 - y^2 - z^2\right)}{6} - \frac{\sqrt{15}G_y xz\left(x^2 - 6y^2 + z^2\right)}{6} - \frac{\sqrt{15}G_z xy\left(x^2 + y^2 - 6z^2\right)}{6}$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{3,3}^{(1,-1)}[g](T_{1g})$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2 y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2 z}{2} - \frac{3y^2 z}{2} + z^3$$

** expression

$$\frac{\sqrt{15}G_x\left(2x^2 - y^2 - z^2\right)}{10} - \frac{\sqrt{15}G_y xy}{5} - \frac{\sqrt{15}G_z xz}{5}$$

$$-\frac{\sqrt{15}G_x xy}{5} - \frac{\sqrt{15}G_y\left(x^2 - 2y^2 + z^2\right)}{10} - \frac{\sqrt{15}G_z yz}{5}$$

$$-\frac{\sqrt{15}G_x xz}{5} - \frac{\sqrt{15}G_y yz}{5} - \frac{\sqrt{15}G_z\left(x^2 + y^2 - 2z^2\right)}{10}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](T_{1g})$, $\vec{\mathbb{G}}_{3,3}^{(1,1)}[g](T_{1g})$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{G_x\left(8x^4 - 24x^2y^2 - 24x^2z^2 + 3y^4 + 6y^2z^2 + 3z^4\right)}{12} + \frac{5G_yxy\left(4x^2 - 3y^2 - 3z^2\right)}{12} + \frac{5G_zxz\left(4x^2 - 3y^2 - 3z^2\right)}{12}$$

$$-\frac{5G_xxy\left(3x^2 - 4y^2 + 3z^2\right)}{12} + \frac{G_y\left(3x^4 - 24x^2y^2 + 6x^2z^2 + 8y^4 - 24y^2z^2 + 3z^4\right)}{12} - \frac{5G_zyz\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$

$$-\frac{5G_xxz\left(3x^2 + 3y^2 - 4z^2\right)}{12} - \frac{5G_yyz\left(3x^2 + 3y^2 - 4z^2\right)}{12} + \frac{G_z\left(3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](T_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](T_{2g})$, $\vec{\mathbb{G}}_{3,3}^{(1,-1)}[g](T_{2g})$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_x\left(y - z\right)\left(y + z\right)}{2} + G_yxy - G_zxz$$

$$-G_xxy - \frac{G_y\left(x - z\right)\left(x + z\right)}{2} + G_zyz$$

$$G_xxz - G_yyz + \frac{G_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](T_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](T_{2g})$, $\vec{\mathbb{G}}_{3,3}^{(1,1)}[g](T_{2g})$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}G_x\left(y - z\right)\left(y + z\right)\left(6x^2 - y^2 - z^2\right)}{12} - \frac{\sqrt{15}G_yxy\left(2x^2 - 5y^2 + 9z^2\right)}{12} + \frac{\sqrt{15}G_zxz\left(2x^2 + 9y^2 - 5z^2\right)}{12}$$

$$-\frac{\sqrt{15}G_xxy\left(5x^2 - 2y^2 - 9z^2\right)}{12} + \frac{\sqrt{15}G_y\left(x - z\right)\left(x + z\right)\left(x^2 - 6y^2 + z^2\right)}{12} - \frac{\sqrt{15}G_zyz\left(9x^2 + 2y^2 - 5z^2\right)}{12}$$

$$\frac{\sqrt{15}G_xxz\left(5x^2 - 9y^2 - 2z^2\right)}{12} + \frac{\sqrt{15}G_yyz\left(9x^2 - 5y^2 + 2z^2\right)}{12} - \frac{\sqrt{15}G_z\left(x - y\right)\left(x + y\right)\left(x^2 + y^2 - 6z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_{2u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}G_x x\,(y-z)\,(y+z)}{2} - \frac{\sqrt{5}G_y y\,(x-z)\,(x+z)}{2} + \frac{\sqrt{5}G_z z\,(x-y)\,(x+y)}{2}$$

$\vec{\mathbb{Q}}^{(1,0)}_{3,1}[g](T_{1u})$, $\vec{\mathbb{Q}}^{(1,0)}_{3,2}[g](T_{1u})$, $\vec{\mathbb{Q}}^{(1,0)}_{3,3}[g](T_{1u})$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{3}G_y z\left(4x^2-y^2-z^2\right)}{4} - \frac{\sqrt{3}G_z y\left(4x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{3}G_x z\left(x^2-4y^2+z^2\right)}{4} - \frac{\sqrt{3}G_z x\left(x^2-4y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{3}G_x y\left(x^2+y^2-4z^2\right)}{4} + \frac{\sqrt{3}G_y x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}^{(1,0)}_{3,1}[g](T_{2u})$, $\vec{\mathbb{Q}}^{(1,0)}_{3,2}[g](T_{2u})$, $\vec{\mathbb{Q}}^{(1,0)}_{3,3}[g](T_{2u})$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\sqrt{5}G_x xyz + \frac{\sqrt{5}G_y z\left(2x^2+y^2-z^2\right)}{4} + \frac{\sqrt{5}G_z y\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}G_x z\left(x^2+2y^2-z^2\right)}{4} - \sqrt{5}G_y xyz - \frac{\sqrt{5}G_z x\left(x^2-2y^2-z^2\right)}{4}$$

$$\frac{\sqrt{5}G_x y\left(x^2-y^2+2z^2\right)}{4} - \frac{\sqrt{5}G_y x\left(x^2-y^2-2z^2\right)}{4} - \sqrt{5}G_z xyz$$